6th Semester		
ISCED Code	**Course Code**	**Course Title**
0613	CSE-3631	Operating System
Credit Hours: 3	**Contact Hours:** 3	**Type: Core, Engineering**
Prerequisite: CSE-3521 (Computer Architecture)		
Co-requisite: None		

Course Content:

#	Content	Duration	CLOs
	Mid-Term (30 Marks)		
1	**Principle of Operating Systems**: Principle of operating systems and Operating system structure: Definition of operating system, Different kinds of operating systems (Desktop, Multiprocessor, Distributed, Clustered, Real time, Handheld systems), Operating-System Services, User Operating-System Interface, System Calls, Types of System Calls, System Programs, Operating-System Design and Implementation, Operating-System Structure, Virtual Machines.	**1 week**	CLO1
2	**Process Management**: Concept of process, process scheduling, CPU scheduling algorithms, threads. thread scheduling, inter process communication	**3 weeks**	CLO2
3	**Process Synchronization** :Critical section problem, race condition, hardware support for synchronization, mutex locks, semaphores, monitors	**1 week**	CLO3
	Final Exam: 50 Marks **Group-A (20 Marks)**		
4	**Deadlock:** Deadlock Characterization, Methods for Handling Deadlocks, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, Recovery from Deadlock.	**2 weeks**	CLO2
5	**Memory Management**: Logical vs. Physical Address Space, Memory management unit, memory protection, Memory Allocation Techniques (continuous memory allocation, paging and segmentation), Swapping	**3 weeks**	CLO3
	Group-B (30 Marks)		
6	**Virtual Memory**: The idea and advantage of virtual memory, Demand Paging, Page Replacement Algorithms ,Thrashing, Allocation of frames	**2 weeks**	CLO3

7	**File Systems**: File Concept, Access Methods, Directory Structure, File-System Mounting, File Sharing, File-System Implementation, Directory Implementation and Allocation Methods, disk scheduling algorithms	**1 week**	CLO3
8	**Protection and Security**: Protection, Principles of Protection, Domain of Protection, Access Matrix, Access Control, Revocation of Access Rights, The Security Problem, Program Threats	**1 week**	CLO2